Experiment 6: K-Means Clustering on Iris Dataset

Aim / Objective

To perform clustering using the K-Means algorithm on the Iris dataset and visualize the clusters.

Theory / Introduction

K-Means is an unsupervised machine learning algorithm that groups data points into *k* clusters based on feature similarity.

- It minimizes the sum of squared distances between data points and their assigned cluster's centroid.
- The number of clusters k must be defined beforehand.
- It is iterative and uses methods like Lloyd's algorithm to converge.

K-Means is useful in scenarios where labels are unknown and natural groupings are to be discovered.

Software / Tools Required

- Python
- Pandas
- Matplotlib
- Scikit-learn

Dataset Used

Iris Dataset

About the Dataset

The **Iris dataset** is widely used for clustering and classification tasks. It contains 150 records of 3 flower species (*setosa*, *versicolor*, *virginica*) with 4 numeric features:

- Sepal length
- Sepal width
- Petal length
- Petal width

Since labels are known, we can visually verify clustering performance.

Procedure / Steps

1. Load the Iris dataset
2. Extract features only (ignore labels for unsupervised learning)
3. Apply K-Means with k=3
4. Visualize clusters using matplotlib
5. Evaluate cluster performance with inertia

Python Step-by-Step Code with Explanations

```
# Step 1: Import required libraries
import pandas as pd
from sklearn.datasets import load_iris
from sklearn.cluster import KMeans
import matplotlib.pyplot as plt

# Step 2: Load the dataset
iris = load_iris()
X = iris.data
features = pd.DataFrame(X, columns=iris.feature_names)

# Step 3: Apply KMeans clustering
kmeans = KMeans(n_clusters=3, random_state=42)
kmeans.fit(X)
labels = kmeans.labels_

# Step 4: Visualize the clusters using first two features
plt.scatter(X[:, 0], X[:, 1], c=labels, cmap='viridis')
plt.xlabel('Sepal Length')
plt.ylabel('Sepal Width')
plt.title('K-Means Clustering (k=3)')
plt.grid(True)
```

```
plt.show()

# Step 5: Print inertia (sum of squared distances)

print("Inertia (Within-Cluster Sum of Squares):", kmeans.inertia_)
```

Result / Output

- Data points were clustered into 3 groups.
- A scatter plot shows the clustering based on sepal dimensions.
- Inertia value indicates compactness of clusters.

Conclusion

K-Means successfully identified clusters in the Iris dataset without using label information. The clustering matched the original species quite well, showing the algorithm's ability to discover natural groupings.

Viva Questions (2–3 per experiment)

1. What is the role of inertia in K-Means clustering?
2. How do you choose the optimal value of *k*?
3. What are the limitations of K-Means?